# Oren Sultan | AI Research Scientist | CS PhD Candidate (Final Semester)

### 3+ years of AI research | 7+ years of engineering

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## 1 Education

- **Feb 2023 - Dec 2026:** PhD Candidate (**direct track for excellent students**) in Computer Science at The Hebrew University, Israel. **PhD Advisor: Dafna Shahaf. Also advised by Yossi Adi (Meta FAIR), and Asi Messica (Lightricks Research).**
- **Oct 2020 - Dec 2022:** M.Sc in Computer Science, The Hebrew University, Israel. Advisor: Prof. Dafna Shahaf. **GPA: 96.5/100. Two long papers were accepted to the main conference of EMNLP 2022. Selected to deliver graduation speech.**
  - **May 2022 - Sep 2022: Erasmus+ Exchange Honors Program & Scholarship**, The Technical University of Munich.
- **Oct 2015 - Mar 2019:** B.Sc in Computer Science, The Hebrew University, Israel. **GPA: 90.21/100, Magna Cum Laude**.
  - **Feb 2018 - Jun 2018:** Student Exchange **Honors Program & Scholarship**, The University of Melbourne.

## 2 Honors & Awards, Scholarships & Fellowships

- **2024 - present: Fellowship for outstanding PhD candidates in AI and Data Science**, The Planning & Budgeting Committee (PBC) of Council For Higher Education (CHE), Israel.
- **2022 - present**: **MSc & PhD Scholarship** – Dafna Shahaf Hyadata-Lab, The Hebrew University, Israel.
- **2022: Erasmus+ Scholarship for honors students**, Technical University of Munich, Germany.
- **2019:** B.Sc in Computer Science – **Magna Cum Laude**, The Hebrew University of Jerusalem, Israel.
- **2018: Scholarship for honors students**, The University of Melbourne, Australia.
- **2013: Annual Excellence Tech Project Award**, The Israeli Intelligence Corps, Commander of Unit 8200 (rank: Tat Aluf).
- **2012: Annual Excellence Team Award**, The Israeli Intelligence Corps, Commander of R&D, Unit 8200 (rank: Aluf Mishne).

## 3 Peer-Reviewed Publications

- **2026:** ALI: Action-Language Inference for Reasoning-Action Alignment in Driving Vision-Language-Action Models**. Oren Sultan**, Yotam Intrator, Idan Kligvasser, Regev Cohen, Ehud Rivlin. In review (NAACL 2027).
- **2026:** HaltEval: Evaluating Non-Termination Reasoning in AI Coding Agents on Real-World Repositories**. Oren Sultan**, Julien Vanegue, Yossi Adi, Peter O'Hearn. In review (NAACL 2027).
- **2026:** LLMs versus the Halting Problem: Revisiting Program Termination Prediction**. Oren Sultan**, Jordi Armengol-Estape, Pascal Kesseli, Julien Vanegue, Dafna Shahaf, Yossi Adi, Peter O'Hearn. **EMNLP 2026 Findings**.
- **2025:** CWM: An Open-Weights LLM for Research on Code Generation with World Models. Meta FAIR CodeGen, **Oren Sultan**. Technical report.
- **2025**: A Neuro-Symbolic Approach for Reliable Proof Generation with LLMs**. Oren Sultan**, Eitan Stern, Dafna Shahaf. **EMNLP 2026 Findings**, ISCOL 2025.
- **2024**: Visual Editing with LLM-based Tool Chaining: An Efficient Distillation Approach for Real-Time Applications **. Oren Sultan**, Alex Khasin, Guy Shiran, Asnat Greenstein-Messica, Dafna Shahaf. **EMNLP 2024**.
- **2024**: ParallelPARC: A Scalable Pipeline for Generating Natural-Language Analogies**. Oren Sultan**, Yonatan Bitton, Ron Yosef, Dafna Shahaf. **NAACL 2024**, ISCOL 2024.
- **2022**: Life is a Circus and We are the Clowns: Automatically Finding Analogies between Situations and Processes**. Oren Sultan**, Dafna Shahaf. **EMNLP 2022**.
- **2022**: Breakpoint Transformers for Modeling and Tracking Intermediate Beliefs. Kyle Richardson, Ronen Tamari, **Oren Sultan**, Reut Tsarfaty, Dafna Shahaf, Ashish Sabharwal. **EMNLP 2022**.

# 4 Professional Experience

- **07.26 - present: Research Scientist Intern, Google Research.**
- **03.26 - 06.26: Research Scientist (part-time), Meta Superintelligence Labs, FAIR.**
- **07.25 - 01.26: Research Scientist Intern, Meta Superintelligence Labs, FAIR.**
- **03.24** - **06.25**: **Research Scientist (part-time), Lightricks Research.**
- **10.23 - 12.23: Research Team Leader, Unit 8200 (Captain in Res.).**
- **08.23 - 02.24**: **Research Scientist Intern, Lightricks Research.**
- **03.23 - 04.23**: **Visiting Researcher**, **University of Surrey & Sheffield.**
- **.05.22 - 07.22: NLP Researcher**, The Technical University of Munich – Social Computing Lab.
- **11.21 - Present: MSc & PhD AI / NLP Researcher**, The Hebrew University, Israel. **Advisor:** Prof. Dafna Shahaf.
- **08.19 - 09.20: Backend Software Engineer (Search), Algorithms group, Fiverr.**
- **11.16 - 12.17: Data Engineer Consultant, Madeira Data Solutions.**
- **06.14 - 10.15: Data Engineer, MyThings.**
- **2008 - 2013: MSSQL Data Engineer, Data Engineering Founder & Team Leader – R&D, Unit 8200.**

# 5 Invited Talks

- **Oct 2026**: "ALI: Action-Language Inference for Reasoning-Action Alignment in Autonomous Driving Vision-Language-Action Models", Google Research Conference, Mountain View, California, USA.
- **Apr 2025**: "Visual Editing with LLM-based Tool Chaining", Nvidia & NLP IL Meetup, Israel.
- **Sep 2024**: "Visual Editing with LLM-based Tool Chaining", AI course, Lightricks, Israel.
- **Sep 2024**: "Fine-Tuning Large Language Models (LLMs)", AI course, Lightricks, Israel.
- **Jun 2023:** "Finding Analogies between Situations and Processes", MSc graduation ceremony, Hebrew University, Israel.
- **Mar 2023**: "Analogies of Situations and Processes: Detection, Explanation, Generation", SheffieldNLP, UK.
- **Jan 2023:** "Using LLMs for Natural-Language Analogies Generation", BIU-NLP, Bar Ilan University, Israel.
- **Jan 2023:** "Automatically Finding Analogies between Situations and Processes", The Hebrew University, Israel.
- **Jan 2022:** "Breakpoint Transformers for Modeling and Tracking Intermediate Beliefs", Aristo team, AI2, Seattle, USA.

# 6 Academic Activities

- **TA:** "self-paced introduction to data science" (67876) at the Center for Interdisciplinary Data-Science Research (CIDR), The Hebrew University, Israel. 2023-2024.
- **Conference Reviewer:** ACL 24, ACL 25, ACL 26, EMNLP 26, AAAI 2027

# 7 Miscellaneous

- **Citizenships**: Israeli, Portuguese.
- **Languages**: Hebrew (mother tongue), English (fluent), Spanish (intermediate), German (elementary).